

Complete Technical & Process Guide

Document Scope

Topic	Scope	Target Release
GxP Compliance	Regulatory framework — 21 CFR Part 11 / EU Annex 11	Ongoing
CGTO / ATO Fields	Industry Object Extension on PO (Therapy, Study, Protocol, Patient)	2702
Change Document	Field-level audit trail for PO (CDHDR/CDPOS)	Done 2608 ✓
eSignature	Electronic signature on regulated PO changes via BTP SaaS	2702
Reauthentication	Platform-level reusable framework — LOB callback integration	2702

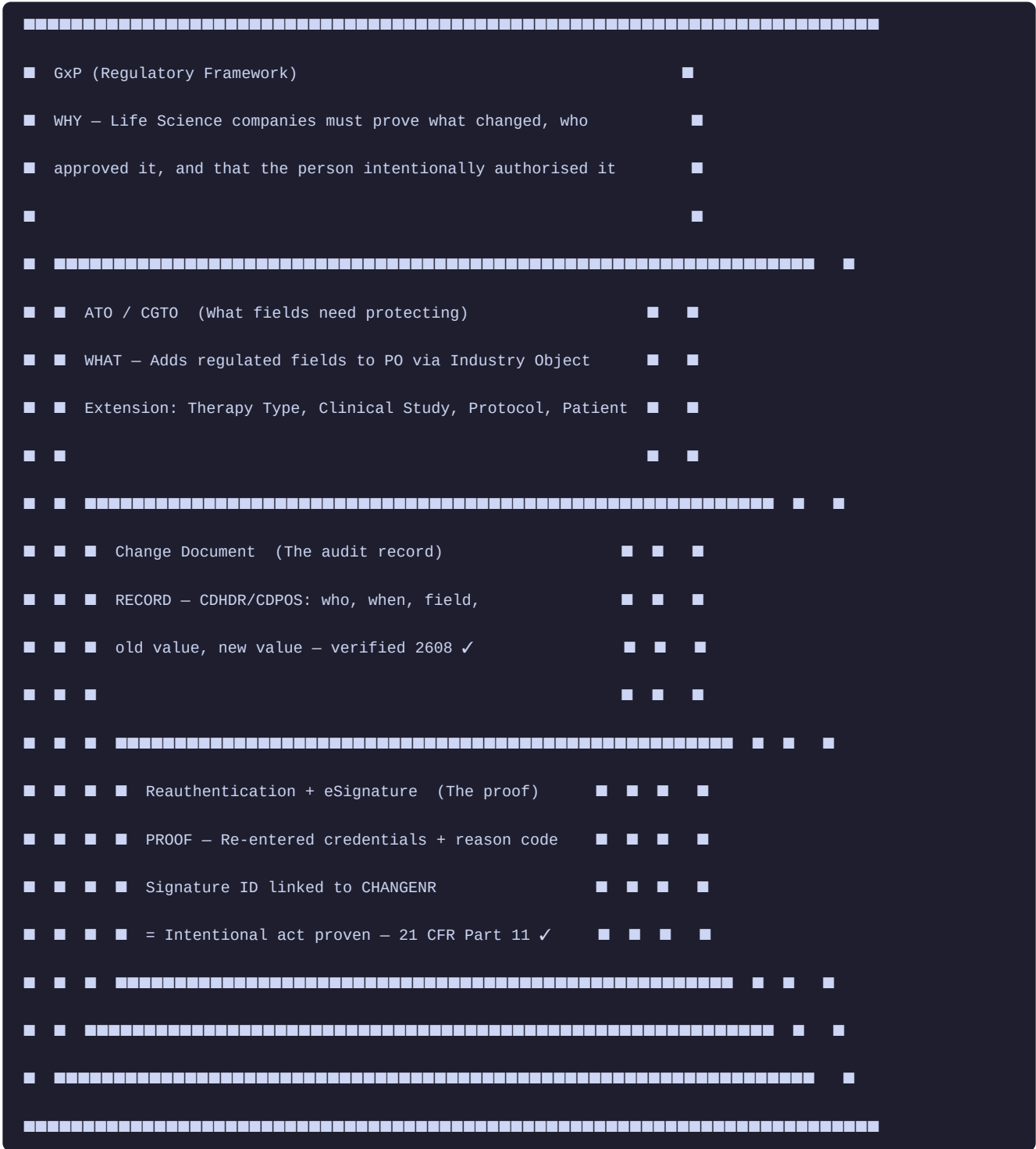
Key Contacts

Name	Role	Work Stream
Swarnava Chatterjee	eSignature Integration Lead	WS1 — eSignature
Yadesh Gupta	Reauthentication Framework Architect	WS1 — Reauth
Prabir Kumar Mallick	Reauth Framework Dev Lead	WS1 — Reauth
Satish Kumar (Meenakshisundaram)	ATO/CGTO PM + Architecture	WS3 — ATO/CGTO
Viswanath Natesan	ATO Architecture	WS3 — ATO/CGTO
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Nils Hartmann	Purchase Order Product Owner	Both
Felix Vente	Suite Architecture — North Star Governance	Governance

Section 1 — How GxP, ATO/CGTO, eSignature & Reauthentication Relate

1.1 The Four Pillars — One Compliance Goal

GxP compliance for Purchase Order in SAP S/4HANA is delivered through four interconnected layers. Each layer depends on the one below it. Removing any layer breaks the compliance chain.



1.2 Dependency Chain

Layer	Delivers	Depends On	Status
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GxP Regulatory	The obligation to comply	—	Ongoing
ATO/CGTO Fields	Regulated fields on PO (Therapy, Study etc.)	Life Science Add-on installed	Dev started 2026
Change Document	Field-level audit trail CDHDR/CDPOS	Standard S4 framework	Done 2608 ✓
Reauthentication / eSignature	Proof of intentional approval linked to change doc	Change Document must work first	Planned 2702

1.3 What eSignature Actually Captures

■ Common Misconception: eSignature is NOT a handwritten signature image or PDF certificate. It is NOT simply recording who was logged in. It is cryptographic proof that the user RE-ENTERED their credentials at the exact moment of the specific change and accepted responsibility for it with a reason code. This distinction is what 21 CFR Part 11 requires.

Aspect	Normal Audit Log	GxP eSignature
What is recorded	User ID + timestamp of save	User ID + timestamp + re-entered credential hash + reason code
Proof of intent	None — user may have walked away	Yes — user actively re-authenticated at that moment
Stored where	S4 change document (CDHDR)	BTP eSignature SaaS (linked to CHANGENR)
21 CFR Part 11	Partial — audit trail only	Full compliance — signature + audit trail
Reason code	Not captured	Mandatory — e.g. Clinical Protocol Update

Section 2 — ATO Extension Tables in S4

2.1 How the Tables Are Delivered

The ATO extension tables are NOT created manually by the LOB team or customer. They are delivered as part of the Life Science Add-on installation by SAP Cloud Operations. Once installed, the tables exist in S4 and the PurchaseOrder OData API is automatically extended to expose these fields.

Life Science Add-on Installation delivers:

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/AT0/S4_P0_HEADER_EXT  (Extension on EKK0 – P0 Header)
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■ Key: EBELN CHAR 10 (→ EKKO.EBELN, FK) ■

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■ Fields: ■
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■	THERAPY_TYPE	CHAR 20	Therapy classification■
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■ CLINICAL_STUDY CHAR 40 Clinical study ref ■

■	PROTOCOL_REF	CHAR 40	Protocol reference	■
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/AT0/S4_PO_ITEM_EXT (Extension on EKPO – PO Item)

■ Key: EBELN CHAR 10 (→ EKPO.EBELN) ■

■ EBELP NUMC 5 (→ EKPO.EBELP, FK) ■

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```
■ Fields: ■
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■ THERAPY_TYPE CHAR 20 Per-item therapy type ■

■ PATIENT_REF CHAR 40 Patient ID (autolog.) ■

Also delivered:

- ✓ Industry Object Extension node registration on EKK0/EKPO

- ✓ PurchaseOrder OData API extended – TherapyType, ClinicalStudy etc.

- ✓ Change Document registration for ATO extension fields

2.2 Why NOT Append Directly to EKKO/EKPO

Approach	Status	Reason
Append ZZTHERAPY directly to EKKO	■ Rejected	Pollutes standard table, blocked by North Star governance
Industry Object Extension node	■ Approved	Clean separation, approved by Suite Architecture (Felix Vente)

2.3 Where ATO Field Changes Are Stored

Storage Location	What It Holds	Purpose
/ATO/S4_PO_HEADER_EXT /ATO/S4_PO_ITEM_EXT	Current live values only (like EKKO)	Latest state — what the PO contains now
CDHDR (Change Doc Header)	OBJECTCLAS, OBJECTID, CHANGENR, USERNAME, UDATE, UTIME	Who changed what PO and when
CDPOS (Change Doc Item)	TABNAME=/ATO/S4_PO_HEADER_EXT, FNAME, VALUE_OLD, VALUE_NEW	Field-level history — old and new values per change
BTP eSignature SaaS	Signature ID, User, Timestamp, Reason Code, Credential hash	Proof of intentional authorisation — linked to CHANGENR

Section 3 — eSignature Integration: When and How It Is Initiated

3.1 Exact Trigger Point Inside ME21N Save

The eSignature trigger point is inside ME21N save processing — AFTER EKKO/EKPO are written but BEFORE the Logical Unit of Work (LUW) is committed to the database. If reauthentication fails or the user cancels, the entire save is rolled back.

ME21N SAVE PROCESSING – TECHNICAL SEQUENCE

PHASE 1 – Standard PO Processing (existing, unchanged)

User clicks SAVE in ME21N



BAPI_PO_CREATE1 / ME_PROCESS_PO_CUST called internally



EKKO written (PO Header – standard)

EKPO written (PO Line Items – standard)

/ATO/S4_PO_HEADER_EXT written (ATO therapy/study/protocol)

/ATO/S4_PO_ITEM_EXT written (ATO therapy/patient ref)



Change Document Framework fires (CHANGEDOCUMENT_CLOSE)

→ CDHDR row created: USERNAME, UDATE, UTIME, OBJECTID

→ CDPOS rows created: one per changed field with VALUE_OLD/VALUE_NEW

Includes /ATO/S4_PO_HEADER_EXT fields if registered

PHASE 2 – Reauthentication Check (NEW – added in 2702)



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Framework calls Relevancy Check callback: ZCL_P0_REAUTH
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PHASE 3 – BTP Reauthentication Dialog (NEW – 2702)

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existing session token is NOT sufficient)



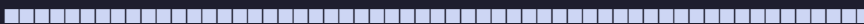
BTP creates Signature Record:

SIG-ID, User, Timestamp, Reason, Credential

S4 calls VALIDATE_RESULT API

Signature ID linked to CHANGENR in CDHDR

PHASE 4 – LUW Commit



COMMIT WORK – atomic:

EKKO / EKPO / ATO extension tables committed

CDHDR / CDPOS committed

Signature ID ↔ CHANGENR link committed

Outbound event published to Event Mesh

→ ATO SaaS notified of PO change in real-time

3.2 The 4 APIs Applications Must Call

API	When to Call	Technology Constraint
REQUEST_REAUTH (OData variant)	Trigger reauthentication dialog	OData / HCP GUI only
REQUEST_REAUTH (library variant)	Trigger reauthentication dialog	All other technologies
VALIDATE_RESULT	Confirm signature was captured successfully	Any technology
CHECK_ESIG_ACTIVE	Check if GxP/eSignature is active — returns true/false	Any technology — call before any signature flow

3.3 Communication Arrangement Setup

A Communication Arrangement must be configured in S4 pointing to the BTP eSignature SaaS tenant. Pattern follows SAP_COM_0267 for other BTP integrations. Requires OAuth / service key exchange between S4 and BTP. This is a one-time setup by the Basis/admin team before any eSignature flow can work.

Section 4 — PO Creation: Complete Sequence (Setup → Config → Process → Runtime)

4.1 Setup Layer — One-Time SAP/Infrastructure

Step	Activity	Performed By	What It Delivers
1.1	Life Science Add-on installed on S4 Private Cloud	SAP Cloud Operations	/ATO extension tables, OData extensions, CD registration
1.2	BTP tenant provisioned	SAP Cloud Operations	ATO SaaS, eSignature SaaS, Event Mesh, iFlow deployed
1.3	Communication Arrangement configured in S4	Basis / Admin	S4 ↔ BTP eSignature OAuth trust, outbound Event Mesh binding

4.2 Configuration Layer — One-Time Functional Setup

Step	SPRO / BTP Activity	Performed By	Result
2.1	Activate scope item 84P — GxP Compliance	Functional Configurator	Reauthentication framework active, S-table entries loaded
2.2	Configure VC_SRA_SOT — Reauth object + class + nodes	GxP Compliance Team	PURCHASE_ORDER registered with callback class and node mapping
2.3	Configure regulated field list per node	GxP Compliance Team	THERAPY_TYPE, CLINICAL_STUDY, PROTOCOL_REF, PATIENT_REF set as triggers
2.4	Define reason codes in BTP for PURCHASE_ORDER	GxP Compliance Team	Clinical Protocol Update, Regulatory Submission etc. available in dropdown
2.5	ATO SaaS master data setup	ATO Configuration Team	Therapy types, clinical studies, protocols, patient numbering defined

4.3 Process Start — Where the PO Journey Begins

From a business/process perspective, the PO journey does NOT start in S4. It starts in ATO SaaS on BTP when a patient is enrolled in a clinical study and the therapy orchestration engine determines that materials need to be procured.

PROCESS START — OUTSIDE S4, IN ATO SaaS (BTP)

Step 0 — Clinical Demand (ATO SaaS)

Patient enrolled in clinical study CS-2026-001

Therapy: Cell & Gene Therapy

Protocol: PROT-47B

Patient Reference: PAT-00123 (autologous therapy)

ATO orchestration engine:

→ "A Purchase Order must be created in S4 to procure materials"

Step 1 – ATO creates PO in S4 via OData API

ATO SaaS → OData call → S4 PurchaseOrder API

Payload:

Standard fields: Vendor 1000234, Material MAT-CGTH-001

Quantity 1 unit, Delivery 2026-08-15

ATO ext fields: TherapyType=CELL_GENE

ClinicalStudy=CS-2026-001

ProtocolRef=PROT-47B

PatientRef=PAT-00123

S4 processes:

EKKO row created

EKPO row created

/ATO/S4_PO_HEADER_EXT row created

/ATO/S4_PO_ITEM_EXT row created

Change Document created (system-to-system – NO reauthentication)

■ GxP rule: automated system-context calls are excluded

Step 2 – S4 confirms back to ATO

Outbound event: PurchaseOrder.Created

→ Event Mesh → iFlow (schema mapping) → ATO SaaS

→ ATO records PO 4500000123 against study CS-2026-001

Step 3 – Human User changes PO in ME21N

Procurement officer opens PO 4500000123

Changes THERAPY_TYPE: PLASMA → CELL_GENE

Clicks SAVE

→ Reauthentication triggered (human user = regulated change)

→ Full Phase 1-4 save flow executes (see Section 3)

Step 4 – ATO SaaS notified of change

Outbound event: PurchaseOrder.Changed

→ Event Mesh → iFlow → ATO SaaS

→ ATO updates therapy orchestration plan in real-time

Step 5 – GxP Audit Trail Complete

CDPOS: THERAPY_TYPE changed PLASMA → CELL_GENE by I308878 at 14:30:22

BTP: SIG-20260624-00891 – I308878 re-authenticated, Reason: Clinical Protocol Update

Combined = 21 CFR Part 11 compliant audit trail ✓

Section 5 — Reauthentication Framework: Configuration & Behaviour

5.1 Three-Level Configuration Model

Level	Location	What Is Configured	Delivered By
Level 1 — Object Registration	S4 S-table VC_SRA_SOT (SPRO)	Reauthentication object name, callback class, authorization class	LOB team via 84P transport
Level 2 — Field Mapping	S4 S-table per node type	Which fields on which node trigger reauthentication	LOB team via 84P transport — customer can fine-tune
Level 3 — Reason Codes	BTP eSignature SaaS	Reason codes attached to each business object type	Auto-provisioned on 84P activation — customer can extend

5.2 Relevancy Check — The Most Critical Method

The Relevancy Check callback is implemented by the LOB application team. It is the single decision point that determines whether reauthentication fires or not. If it returns FALSE — even due to a bug — reauthentication is silently skipped.

Scenario	Return Value	Framework Action
No regulated field changed in this save	ABAP_FALSE	Skip reauthentication entirely — LUW commits normally
A configured regulated field changed	ABAP_TRUE	Trigger reauthentication dialog — LUW paused until signed
Exception raised / error in callback	Treated as FALSE	■ Reauthentication skipped — application bug risk

5.3 Mass Update & Background Job Behaviour

Scenario	Reauthentication Behaviour
User selects 10 objects, changes a regulated field	Single dialog — user signs ONCE — signature covers all 10 objects
Background job run by system/technical user	EXCLUDED — GxP rule: automated system-context jobs do not require signature
Background job triggered by user from Fiori UI	Collect signature ONCE at trigger point (button click) — covers all resulting objects
Scheduled job (independent background process)	Out of scope — must run as system user

5.4 Current Limitation — Value-Level Triggering

Current capability: reauthentication triggers when a configured field changes TO ANY VALUE. Not yet supported: trigger only when field changes from value A to specific value B (e.g. trigger only when status changes to REL, not for all status changes). This is on the framework roadmap but NOT delivered in 2702. Confirmed by Yadesh Gupta: 'We do not have that capability as of now — it will be delivered later.'

Section 6 — Delivery Timeline & Open Items

6.1 Release Timeline

Release	Deliverable	Status
2025	GxP NOT enabled in S/4HANA Cloud	Historical
2602 (2026)	Discussions started, POC work, architecture decisions	Done
2608 (2026)	Change Document for PO verified working for all relevant fields	✓ Done
2702 (2027)	eSignature in ME21N + ATO/CGTO field extension + Reauthentication framework	Planned
Post-2702	eSignature in modernised Manage PO App (BOF replacement)	Deferred — pending app modernisation
Future	eSignature in new RAP-based modern PO UI	Future

6.2 Open Items

#	Open Item	Work Stream	Target
1	eSignature trigger integration in ME21N save flow	WS1	2702
2	Change Document registration for /ATO/S4_PO_HEADER_EXT fields	WS1 + WS3	2702
3	ATO/CGTO field extension OData API exposure	WS3	2702
4	Outbound event publishing S4 → Event Mesh for PO changes	WS3	2702
5	iFlow schema mapping for ATO extension fields	WS3	2702
6	Value-level triggering (field changes FROM value A TO value B)	WS1	Post-2702
7	eSignature in Manage PO App (BOF/Fiori)	WS1	Deferred
8	Mass update background job reauthentication handling	WS1	Under review

6.3 Integration Checklist for LOB Application

Backend / ABAP

- Implement Reauthentication callback class (Relevancy Check)
- Implement Authorization class
- Register reauthentication object in S-table VC_SRA_SOT
- Define node types and field mappings
- Configure regulated fields per node type
- Add content delivery to scope item 84P transport

BTP Setup

- Register reauthentication object name in BTP
- Define / attach reason codes to the object
- Verify reason codes appear for correct object only

Application Code

- Call REQUEST_REAUTH at correct trigger point (before save / on button press)
- Call VALIDATE_RESULT to confirm signature captured
- Call CHECK_ESIG_ACTIVE before any signature flow
- Handle ABAP_FALSE from relevancy check — no silent skip on error

Testing

- Test with scope item 84P active
- Test relevancy check returns TRUE only for regulated field changes
- Test dialog shows correct reason codes
- Test signature is linked to change document CHANGENR
- Test mass update — single sign covers all objects
- Test background job (system user) — reauthentication must be skipped